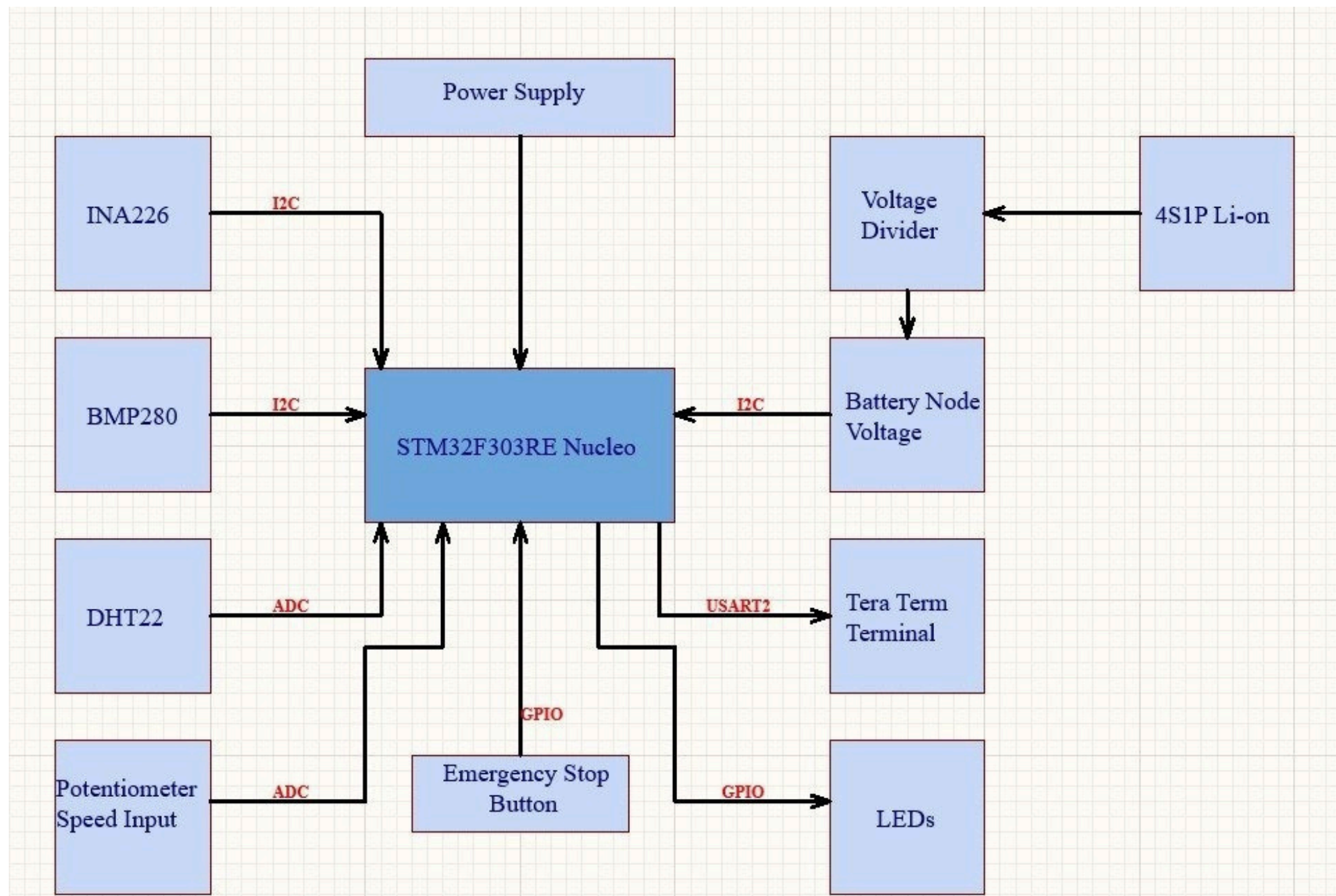




Problem & Motivation

Conventional battery management systems (BMS) halt operations completely during minor sensor failures or I²C bus lockups, causing critical system downtime. This project implements a fault-tolerant, bare-metal BMS with hardware-level bus recovery (I²C defibrillator) and sensor fallback mechanisms to ensure uninterrupted, safe operation.

System Architecture



Real-World Comparison

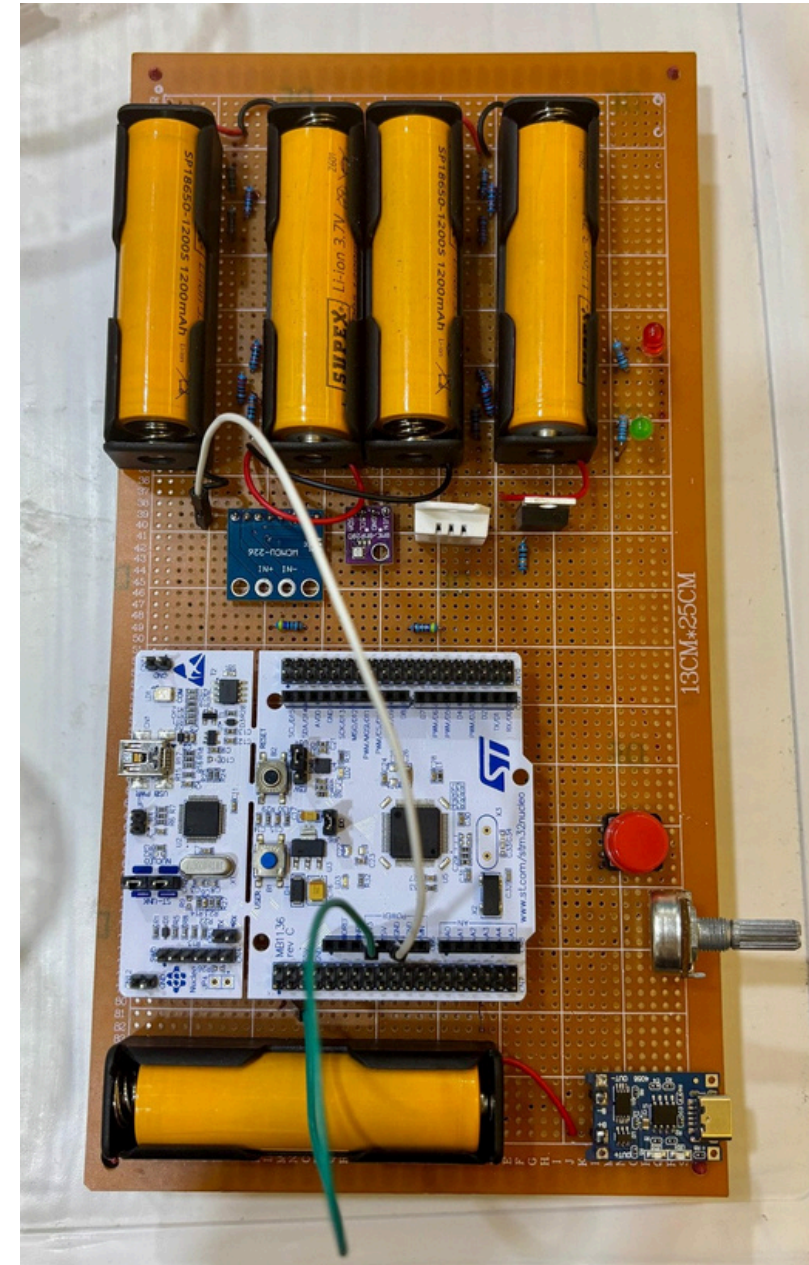
Feature	This Project	Commercial Smart BMS
Voltage, Current & SoC Monitoring	✓	✓
Telemetry & Data Output	✓ (1 Hz UART Real-Time)	✓ (Bluetooth Delayed)
I ² C Bus Lockup Recovery	✓ (Hardware Defibrillator)	✗ (System Halts)
Sensor Fallback Mechanism	✓ (ADC Fallback)	✗ (Cuts Output)
Soft-Start Motor Control	✓ (Dynamic PWM)	✗ (Binary On/Off)
Firmware Architecture	Bare-Metal STM32 (Custom)	Proprietary (Closed)
Approximate Cost (BOM)	~₺ 850	~₺ 1500-3000

While commercial smart BMS provide basic protection, they lack fault-tolerance — halting during sensor dropouts. This project ensures uninterrupted operation at a fraction of the cost.

Methodology

- I²C Bus Recovery (Defibrillator):** At startup and during faults, 9 clock pulses are sent on SCL to recover locked slave devices (INA226/BME280).
- Sensor Fallback Algorithm:** When INA226 communication is lost, the system automatically switches to ADC battery node voltage data.
- Soft-Start Motor Control:** Time-dependent PWM ramp-up algorithm prevents high inrush discharge currents during motor startup.
- Float-to-Int String Formatting:** Decimal numbers split into integer and fractional parts via manual string concatenation.
- Bare-Metal Firmware:** Direct register-level (CMSIS) configuration instead of HAL for low-latency, high-efficiency operation.

Hardware Photo



Conclusion

- Bare-metal STM32F411 firmware implemented with fault-tolerant BMS architecture.
- I²C 'defibrillator' ensures hardware-level bus recovery from sensor lockups in < 1 ms.
- Sensor fallback guarantees uninterrupted monitoring via ADC backup.
- Soft-Start PWM algorithm prevents inrush current damage during motor startup.
- Prototype demonstrates industrial-grade reliability with low-cost components (~₺ 850 BOM).

Key Results

< 1 ms
I²C Lockup
Recovery

< 0.3 V
Imbalance
Max Cell

~₺ 3000
Estimated
BOM

Stable STM32F411 operation with I²C hardware recovery, sensor fallback, soft-start motor control, and real-time USART2 telemetry at 115200 baud.

References

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- [2] Texas Instruments, INA226 Current & Power Monitor, SBOS547D, 2015.
- [3] Bosch Sensortec, BME280 Humidity & Pressure Sensor, Rev. 1.6, 2018.
- [4] Aosong Electronics, DHT22 (AM2302) Humidity & Temperature Sensor.